ChatGPT

Standards and Research Guide for an IAB-Based MAC/Protocol Simulator with Ambient-IoT Extensions

Executive summary

For a simulator with an **abstracted physical layer** and emphasis on MAC/network behavior, the standards-critical IAB stack is not limited to the six documents in your prompt. The minimum coherent 3GPP set is:

TS 38.300 → TS 38.401 → TS 38.331 / TS 38.321 / TS 38.322 → TS 38.340 → TS 38.470 + TS 38.473 (+ TS 38.474 for F1-U), with **TS 38.423/XnAP** required when you simulate inter-donor/inter-CU topology adaptation. If you model a split CU-CP/CU-UP, current E1 specifications in the **37.48x family** also become relevant. [1]

For the **Tag/CFA/Msg2/D2R** part of your thesis, there is an important correction: **38.391 is a TS, not a TR**. The current Release-19 document is **TS 38.391 V19.4.0, uploaded 2026-09-25**, titled *NR; Ambient IoT Medium Access Control (MAC) protocol*. CFA paging, the A-IoT Msg1/Msg2 exchange, D2R scheduling information, D2R TBS and D2R segmentation are A-IoT mechanisms defined principally there, not IAB mechanisms. [2]

The most consequential standards issue for your current system model is this:

> **[A — normative TS] Release-19 A-IoT does not support the split-gNB architecture.**

TS 38.300 V19.4.0 explicitly states that, for the A-IoT architecture in this version, A-IoT-specific information is carried between the gNB and A-IoT CN node over NG-C, no A-IoT-specific communication occurs between gNBs, and **"split gNB architecture is not supported."** [3]

Your current model places a **"DU / Reader" inside each IAB node**, while the node also contains an IAB-MT and is controlled by a CU; the model further lets that Reader page Tags, provide continuous CW and receive D2R backscatter. Pages 4, 8, 10 and 11 of your system-model deck make this architecture and scheduling relationship explicit. filecite turn0file0

Therefore:

[C — implementation inference] your proposed **IAB-node DU/Reader is best described in the thesis as a research extension that combines a standards-based IAB transport model with standards-based A-IoT MAC behavior. It is not, as of Release 19, a standardized A-IoT-over-IAB split-gNB architecture.**

That distinction is crucial and should appear prominently in your system-model chapter. It does **not** invalidate the research. In fact, it gives you a clearly identifiable research extension; you simply should not state that 3GPP currently mandates an A-IoT reader inside an IAB-DU.

A second major finding concerns your **per-Tag Msg2 TBS decision**. In TS 38.391 Msg2, there is **one 24-bit D2R Scheduling Info field for the Random ID Response PDU as a whole**, followed by one to eight ID entries. The per-ID-entry information contains identity-related information and, conditionally, a frequency index; it does **not** contain a separate D2R Scheduling Info/TBS for every ID entry. [4]

Thus:

[A] a single standard Msg2 carrying several responding Tags does **not define independent per-Tag D2R TBS fields**.

[C] a standards-faithful way to realize your decision variable D_k^{TBS} is to schedule Tags through separate Msg2 PDUs—e.g. one ID entry per Msg2—so each Msg2 can carry its own D2R Scheduling Info. Another clean option is CFA, where the D2R scheduling information is conveyed directly in the Paging message. The feasibility/timing of separate Msg2 transmissions should be treated as a simulator scheduling decision, not as a requirement stated by 3GPP.

A third important distinction concerns your **DU time window**. IAB does have normative DU/MT time-resource restrictions, but 3GPP expresses them as **symbol-level Hard, Soft and Unavailable attributes**, conditional Soft-resource availability, half-duplex constraints and guard symbols. There is no generic normative IAB parameter called `t_DU,start` / `t_DU,end`. [5]

Accordingly, the DU window drawn on page 10 and the constraint

$$t_{j,k}^{D2R,start} + T_{j,k}^{D2R} \leq t_j^{DU,end}$$

on page 11 of your model are well-motivated **[C — abstractions]**, but are not verbatim 3GPP requirements. 🀄filecite🀄turn0file0🀄

For your implementation, I recommend calling the result a **“Release-19-aligned protocol-level IAB/A-IoT simulator with an abstract PHY and an explicitly non-standard IAB-node reader extension”**, rather than claiming formal 3GPP conformance. Formal radio-node conformance would additionally involve PHY/RF specifications such as TS 38.174 and TS 38.176, which are intentionally outside your model. [6]

Standards baseline and compliance boundary

I use the following notation throughout:

[A — Normative] means a requirement or defined behavior from a 3GPP TS.

[B — Research/study] means either a non-normative 3GPP TR or a finding from an IEEE/academic publication.

[C — Inference] means a reasonable simulator design derived from the standards but not itself specified by 3GPP.

Because A-IoT MAC entered the Release-19 specification set, a **Release-19 baseline** is the cleanest target for your combined IAB/A-IoT simulator. The current documents available in October 2026 include TS 38.300 V19.4.0, TS 38.331 V19.4.0, TS 38.423 V19.4.0, TS 38.391 V19.4.0, TS 38.401 V19.3.0 and TS 38.470 V19.2.0. [7]

Comparative specification map

Document	Release-19 version/date used here	Scope	Main requirements relevant to your simulator	Simulator impact
TS 38.300	**19.4.0, 2026-09-25**	Overall NR/NG-RAN Stage-2	IAB architecture; IAB-DU/ IAB-MT; F1 over BAP; resource configuration; A-IoT Stage-2	**Essential:** architecture, scheduler, IAB/A-IoT boundary
TS 38.401	**19.3.0, 2026-06-22**	NG-RAN architecture	CU/DU/IAB roles; F1 relationships; IAB procedures and BH-RLC establishment	**Essential:** entity model and control flow
TS 38.340	**19.0.0, 2025-10-10**	Backhaul Adaptation Protocol	BAP routing, next hop, BH-RLC mapping, header rewriting, flow control, BH-RLF handling	**Essential for multi-hop IAB data plane**
TS 38.331	**19.4.0, 2026-09-27**	NR RRC	IAB-MT configuration; BAP address/default route/default BH-RLC; BH-RLC add/modify/ release; IAB IP config	**Essential control plane**
TS 38.321	**19.4.0, 2026-09-25**	NR MAC	MAC procedures plus IAB timing offsets, guard symbols, beam/power/ timing-control CEs	**Essential scheduling semantics**
TS 38.322	**19.3.0, 2026-06-28**	NR RLC	AM/UM RLC behavior, segmentation/ reassembly/ retransmission; BH-RLC instances	**Essential if queues/ retransmissions are modeled**
TS 38.470	**19.2.0, 2026-06-22**	F1 general aspects	CU-DU F1 framework; IAB BAP mapping/resource/ TNL procedures	**Essential framework**

Document	Release-19 version/date used here	Scope	Main requirements relevant to your simulator	Simulator impact
TS 38.473	**19.4.0, 2026-09-27**	F1AP	Actual F1AP procedures and IEs for routes, BH-RLC mapping, resources and IAB configuration	**Essential message/ API semantics**
TS 38.474	**19.0.0, 2025-10-02**	F1 data transport	F1-U transport	Model as abstract tunnel unless packet-level transport matters
TS 38.423	**19.4.0, 2026-09-27**	XnAP	Inter-CU/inter-donor IAB migration and resource coordination	Conditional: topology migration/multi-donor
TS 38.391	**19.4.0, 2026-09-25**	**A-IoT MAC**	Paging, CFA/CBRA, Msg1/ Msg2, D2R scheduling/ TBS, D2R MAC PDUs, segmentation	**Essential for your Tags; not an IAB protocol**
TS 38.291	**19.3.0, 2026-03-20**	A-IoT PHY	Timing/resource interpretation used by 38.391	Abstract into PHY service contract
TS 38.413	**19.4.0, 2026-09-27**	NGAP	A-IoT Inventory/Report/ Command signaling toward AIOTF/AMF	Required if CN/A-IoT service procedures are modeled
TR 38.874	**16.0.0, 2019-01-10**	Original IAB study	Architectural alternatives, scheduling, QoS, fairness and congestion study	**Non-normative rationale / scheduler research**
TR 38.809	**16.6.0, 2022-04-01**	IAB radio transmission/ reception background	RF/PHY assumptions behind IAB	Low priority for abstract PHY
TR 33.824	**17.0.0, 2022-03-24**	IAB security study	Threat/security study	Optional unless security modeled
TR 28.875	**19.0.0, 2025-01-02**	IAB management study	OAM/management, including newer IAB-management work	Optional for management-plane studies

The version/date entries above come from the official 3GPP specification portal and ETSI's official publication of the corresponding 3GPP specifications. [8]

The documents in your original list therefore are **not the full relevant set**. In particular, omitting **TS 38.340 (BAP)** and **TS 38.473 (F1AP)** would leave out two of the most important specifications for an actual multi-hop IAB simulator. TS 38.300 explicitly places IP/F1 traffic over BAP on the wireless backhaul, while TS 38.340 defines BAP's routing and BH-RLC mapping behavior. [9]

Official source pages for the primary documents:

Specification	Official 3GPP source
TS 38.300	3GPP portal — 38.300
TS 38.391	3GPP portal — 38.391
TS 38.331	3GPP portal — 38.331
TS 38.423	3GPP portal — 38.423
TS 38.401	3GPP portal — 38.401
TS 38.470	3GPP portal — 38.470
TS 38.473	3GPP portal — 38.473
TS 38.340	3GPP portal — 38.340
TS 38.321	3GPP portal — 38.321
TS 38.322	3GPP portal — 38.322
TS 38.323	3GPP portal — 38.323
TS 38.474	3GPP portal — 38.474
TS 38.291	3GPP portal — 38.291
TS 38.413	3GPP portal — 38.413

Specification-by-specification implementation guide

Core architecture, BAP, RRC and MAC

TS 38.300 — NR and NG-RAN Overall Description, Stage-2

Purpose. This is the architectural starting point. Clause **4.7** defines IAB architecture and protocol stacks; **6.11** describes BAP-layer traffic mapping/routing behavior at Stage-2 level; **10.9** defines IAB resource configuration; **16.23** describes Release-19 A-IoT. [10]

[A] Key requirements.

- An IAB node contains gNB-DU functionality toward UEs/child IAB nodes and uses its IAB-MT toward its parent. An IAB donor is the topology root, and the donor performs centralized resource, topology and route management. [11]

- F1-C and F1-U between the IAB-DU and donor CU use IP; across the wireless backhaul that IP is transported over **BAP**, with hop-by-hop BH-RLC underneath. [12]
- The IAB-DU scheduler must obey its configured IAB resources. Each symbol is classified **Hard, Soft or Unavailable**; Soft symbols are usable conditionally, including by explicit parent indication using DCI format 2_5 or by the specified implicit determination. Guard symbols may be used at IAB-MT/IAB-DU transitions. [5]
- A-IoT is a separate radio stack consisting of A-IoT MAC and A-IoT PHY, without the ordinary CP/UP distinction on that radio interface. D2R uses backscatter on a single-tone carrier wave. [13]
- Release-19 A-IoT does not support a split-gNB architecture. [3]

Optional/capability-dependent IAB features: enhanced spatial/frequency multiplexing, topology redundancy, migration, mobile IAB, alternative routing and enhanced timing modes. The enhanced multiplexing cases permit combinations such as simultaneous MT-Rx/DU-Rx or MT-Tx/DU-Tx when the node has the required capability; these should not be assumed for a baseline half-duplex node. [5]

Simulator APIs/classes: `IabDonor`, `IabNode{MT,DU}`, `ParentLink`, `ChildLink`, `ResourceGrid`, `SymbolAvailability`, `GuardPeriod`, `RouteManager`, and a PHY-independent callback such as `isDuSchedulable(symbol)`.

For your thesis, clauses **4.7.1–4.7.4, 6.11.1–6.11.3, 10.9, 16.23.1–16.23.5** are especially important. [10]

TS 38.401 — NG-RAN Architecture Description

Purpose. 38.401 refines the functional entities and interface relationships introduced by 38.300. The Release-19 version used here is V19.3.0. [14]

[A] Key requirements. The IAB donor comprises CU and donor-DU functionality; an IAB node has an IAB-DU and an IAB-MT. The document's IAB procedures in clause **8.9** cover integration and backhaul functions including BH-RLC establishment, traffic mapping, release, OAM and IP address allocation. [15]

Of particular value to a simulator is **8.9.8**, which gives a concrete BH-RLC establishment signaling flow involving CU↔DU F1AP UE context modification and RRCReconfiguration toward the IAB-MT. [16]

Simulator APIs: topology/entity ownership, CU↔DU configuration, `createBhRlcChannel()`, `deleteBhRlcChannel()`, IP address allocation and a state machine for IAB-node integration.

Reference clauses: 6.1.3 for IAB architecture and **8.9.x**, especially **8.9.8–8.9.13**, for operational procedures.

TS 38.340 — Backhaul Adaptation Protocol

This is arguably the most important additional specification missing from your initial list. The current Release-19 version is V19.0.0. [17]

[A] BAP functions include data transfer, determining destination/path, selecting egress BH-RLC channels, forwarding to the next hop, BAP-header rewriting, distinguishing locally terminated versus forwarded traffic, flow-control signaling and handling BH-RLF indications. [18]

Routing is performed using a configuration derived from F1AP. A route includes a **BAP routing ID**, whose routing semantics include a BAP address/path identity and next-hop information. [19]

BAP also supports hop-by-hop flow-control feedback; an IAB-DU may regard a BH link as congested for a routing ID when the received available-buffer indication falls below the configured threshold. [20]

Simulator API:

```
BapEntity
  route(pdu, routingId) -> nextHop
  mapToBhRlc(pdu, nextHop, trafficClass) -> BhRlcChannel
  rewriteHeader(...)
  receiveFlowControl(...)
  reportFlowControl(...)
  markLinkRlf(...)
  localReroute(...)
```

For a multi-hop simulator, BAP should be an actual logical layer rather than a direct donor-to-node “wireless pipe”. Otherwise important IAB phenomena—route-dependent queueing, per-hop congestion and BH-RLC differentiation—disappear.

Reference clauses: 4.4–4.5, 5.2 routing/mapping, 5.3 flow control, 6 BAP PDU formats. [21]

TS 38.331 — RRC

The current Release-19 document is V19.4.0. [22]

[A] IAB-relevant RRC configuration includes `bap-Config`. RRC can establish a BAP entity, configure the node BAP address, configure the default uplink BAP routing ID, select the default UL BH-RLC channel and configure the type of BAP flow-control feedback. [23]

The ASN.1 `BAP-Config-r16` includes a 10-bit BAP address, `defaultUL-BAP-RoutingID`, `defaultUL-BH-RLC-Channel` and flow-control feedback type. The default routing/BH-RLC configuration is important during IAB bootstrap and related recovery/migration situations. [24]

Simulator APIs: RRC messages do not need bit-perfect ASN.1 encoding unless you want interoperability, but their **state changes do need to be represented**:

```
RrcReconfiguration:
    add/modify/release BH-RLC
    set BAP address
    set default UL BAP route
    set default UL BH-RLC
```

```
    configure flow-control feedback
    configure IAB IP address information
```

Reference clauses: 5.3.5, especially the IAB-specific BH-RLC/BAP/IP subsections including **5.3.5.12a**, plus the ASN.1 definitions of `BAP-Config-r16`. [23]

TS 38.321 — NR MAC

The current Release-19 version is V19.4.0. [25]

For an abstract PHY, implement standard MAC scheduling state and the IAB-specific control effects rather than OFDM processing. Particularly relevant clauses include **5.18.18 Timing offset adjustments for IAB**, **5.18.19 Guard symbols for IAB**, **5.18.26 restricted/recommended beam indication**, **5.18.27 DL Tx-power adjustment**, **5.18.28 UL PSD range adjustment** and **5.18.29 timing-case indication**. [26]

[C] For your simulator, beam and power-control CEs can be represented as metadata/events if your objective does not require RF modeling. Timing and resource-availability effects should not be abstracted away, because they directly determine how much DU airtime remains for UE/Tag service.

F1, Xn and E1 interfaces

TS 38.470 — F1 general aspects and principles

V19.2.0 was published in July 2026. [27]

[A] F1 is the CU↔DU interface used by the IAB-DU to terminate at the donor CU. Its IAB support encompasses BAP mapping/configuration, IAB DU resource configuration and IAB transport-network information. For your simulator, 38.470 defines the conceptual interface, while **38.473 is where the actual F1AP messages and IEs live**.

Implement at minimum an abstract:

```
F1ControlService:
    f1Setup()
    ueContextSetup/Modify/Release()
    configureIabDuResources()
    configureBapRouting()
    configureBhRlcMapping()
    configureTransportAddress()
```

TS 38.473 — F1AP

V19.4.0 was uploaded on 2026-09-27. [28]

This is essential even if F1 is not encoded on a real SCTP socket. It carries configuration needed to make the IAB Stage-2 behavior operational, including BAP routing/mapping, BH-RLC identifiers, IAB-DU/cell resource information and related IAB configuration. An official V19.2.0 publication, for example, contains IAB-specific information in the F1AP IE set, while the portal shows the current V19.4.0 version. [29]

[C] For a thesis simulator, use typed structs/events corresponding to F1AP information elements rather than ASN.1 PER encoding. ASN.1 encoding is necessary only if you intend to interoperate with a real CU/DU implementation.

TS 38.474 — F1 data transport

V19.0.0 is the Release-19 publication. [30]

[C] With an abstract PHY/network implementation you can model F1-U as a logical IP/GTP-U flow with byte count, QoS class and latency/queue state. You do not need a real UDP/GTP-U socket unless transport-stack realism is part of the experiment.

TS 38.423 — XnAP

Current Release-19 version: **V19.4.0, September 2026**. [31]

XnAP includes IAB-specific procedures for inter-donor/inter-CU cases, including IAB transport migration management and modification. Even earlier normative releases show `IAB TRANSPORT MIGRATION MANAGEMENT` message families whose purpose is to coordinate traffic migration between F1-terminating and non-F1-terminating IAB donors. [32]

Implementation rule: if your topology is one static donor and no IAB-node migration crosses donor/CU boundaries, **XnAP can be out of the minimum executable profile**. If you test inter-donor migration or topology adaptation, it must enter the simulator.

E1. E1 is not an IAB-specific protocol; it is needed only if you explicitly split the CU into CU-CP and CU-UP. For a current Release-19 implementation, the E1 specifications are in the **37.48x family**, including TS 37.480 and TS 37.483, after extension of the older 38.46x family. [33]

[C] A single logical CU object is therefore the right minimum for your thesis unless CU-CP/CU-UP effects are themselves research variables.

A-IoT MAC, CFA, Msg2 and D2R

TS 38.391 — Ambient IoT MAC

Official source: 3GPP TS 38.391. It is **TS 38.391 V19.4.0**, not “TR 38.391”. [34]

The clauses most relevant to your model are:

- **5.2** Paging

- **5.3.1** contention-based access
- **5.3.2** contention-free access
- **5.4.2** D2R message transmission
- **5.4.4** D2R segmentation
- **6.2.1.1** A-IoT Paging message
- **6.2.1.3** Random ID Response / Msg2
- **6.2.1.6** D2R Scheduling Info
- **6.2.2.x** D2R message formats. [35]

[A] CFA. When CFA is selected by Paging, the MAC entity proceeds directly to the D2R transmission procedure. There is no CBRA Msg1/Msg2 contention-resolution exchange in this branch. [36]

[A] CBRA. The device transmits a 16-bit random ID in Msg1. After a successful Random ID Response/Msg2 match, the device transmits its D2R upper-layer message in the resources indicated by Msg2. [37]

[A] Msg2 structure. The Random ID Response contains one **24-bit D2R Scheduling Info** followed by a list containing **one to eight ID entries**. The frequency-index-presence indication applies to all ID entries. [38]

This is the critical implication for your thesis:

> **[C] Do not model `D_TBS[k]` as a per-ID-entry field inside one multi-Tag Msg2.**

The standardized message format does not provide that field.

A safe model is:

$$\text{one scheduled Tag} \rightarrow \text{one Msg2 with its own D2R Scheduling Info/TBS}$$

or, for CFA:

$$\text{Tag-specific Paging} \rightarrow \text{Tag-specific D2R Scheduling Info/TBS}.$$

This preserves your per-Tag optimization variable while avoiding the false statement that 3GPP defines "per-Tag TBS inside the ID-entry list."

[A] D2R Scheduling Info. The information controls fields such as bit duration, frequency-resource selection, repetition/coding-related behavior and D2R TBS. The MAC derives the corresponding L1 parameters and D2R TBS from it. [39]

[A] D2R TBS. The D2R-transmission procedure constructs a D2R upper-layer transfer message according to the scheduled TBS; a smaller payload is padded, while a payload too large for the scheduled resource leads to segmentation behavior where applicable. [40]

So your `D_TBS[k]` decision should be represented in the simulator as:

```
AiotD2rGrant {
    tagId
    scheduledTbsBytes
    bitDuration
    frequencyResource
    repetitionConfig
    codingConfig
    other 38.391 scheduling fields
}
```

not merely as a number of bits detached from the standard D2R Scheduling Info.

TS 38.291 — A-IoT Physical Layer

Current Release-19 version on the 3GPP portal is V19.3.0. [41]

Because you abstract PHY, you do not need waveform generation. You do, however, need a **PHY service contract** that accepts the parameters derived from 38.391 and returns transmission duration/success/received bits. TS 38.300 confirms that D2R is backscatter on a single-tone carrier wave and that the PDRCH carries the MAC-originated D2R transport block. [13]

A suitable abstraction is:

```
AiotPhy.txDuration(D2rGrant, tbs)
AiotPhy.canDecodeR2d(tag, reader, channelState)
AiotPhy.canDecodeD2r(tag, reader, channelState, cwAvailable)
AiotPhy.deliverD2rMacPdu(...)
```

[C] This is enough for MAC-level research, provided your paper explicitly states that coding, OOK/BPSK waveform generation, chip mapping and RF impairments are abstracted.

TS 38.413 — NGAP

V19.4.0 contains a dedicated **clause 8.20 Ambient IoT Procedures**. The Inventory Request is a non-UE-associated procedure requesting a gNB to perform A-IoT paging; Inventory Report carries the resulting reports toward the A-IoT CN node. [42]

This interface matters if your service requests S_i, required Tag sets T_i^{req}, deadlines and required data amounts are meant to originate from the standardized A-IoT core-side service flow rather than from an internal experiment generator.

[C] For your current optimization-focused simulator, it is reasonable to represent NGAP as a service-request API rather than instantiate an AMF/AIOTF implementation:

```
InventoryRequest {
    correlationId
    targetArea/readers
    serviceParameters
}

InventoryReport {
    correlationId
    deviceReports[]
}
```

but that abstraction should be documented.

Mapping the standards to your simulator and thesis model

Your deck defines a farm deployment in which IAB provides wireless backhaul, Tags harvest energy from continuous wave transmission and communicate by backscatter, and each IAB-node-side scheduler selects Tags and assigns TBS. It also defines per-service deadlines/minimum data, Tag energy state, accumulated data B_k, a fairness objective and a DU availability interval. filecite turn0file0

The following map cleanly separates standardized state from research state.

Your simulator component / variable	3GPP status	Standards mapping	Recommended implementation
IAB donor CU/DU	**A**	38.300 §4.7; 38.401	Explicit entities
IAB node MT + DU	**A**	38.300 §4.7	Explicit entities
Multi-hop wireless BH	**A**	38.300 + 38.340	BAP + per-hop BH-RLC
BAP routing ID / next hop	**A**	38.340	Route table
BH-RLC channels	**A**	38.331/38.322/38.340	One or more logical queues per hop
F1-C/F1-U	**A**	38.470/473/474	Abstract typed interface/tunnel
Hard/Soft/ Unavailable DU resources	**A**	38.300 §10.9	Symbol availability map
Guard symbols	**A**	38.300 §10.9; 38.321 §5.18.19	Scheduler exclusion intervals

Your simulator component / variable	3GPP status	Standards mapping	Recommended implementation
`t_DU,start`, `t_DU,end`	**C**	Derived from usable symbols	Derived contiguous window, not a standard IE
Tag Paging	**A, A-IoT**	38.391	A-IoT Reader MAC
CFA	**A, A-IoT**	38.391 §5.3.2	Direct Paging→D2R branch
CBRA Msg1 / Msg2	**A, A-IoT**	38.391 §5.3.1/6.2.1.3	State machine
Per-Tag TBS	**A/C**	TBS itself standardized; per-Tag optimization is your design	One Tag/grant or one Tag-specific CFA paging
D2R backscatter	**A, A-IoT**	38.300 §16.23; 38.291/391	Abstract D2R PHY + MAC
Continuous CW duration	Partly **A/C**	CW/backscatter standardized; your scheduling duration/model is research	PHY/energy model
Battery/EH differential equation	**C/B**	Not defined by IAB MAC specs	Plug-in energy model
$E_k^{bk} + E_k^{EH} \geq E_{min}$	**C**	Thesis feasibility constraint	Optimization constraint
B_k fairness objective	**B/C**	Fair scheduling studied, not mandated	Policy layer
Service deadline T_i^{max}	**C** unless mapped to service/QoS	Application requirement	Deadline scheduler
D_i^{min}	**C**	Application requirement	Completion criterion

The standard's IAB resource model is finer grained than your current rectangular DU window. TS 38.300 requires the DU scheduler to comply with symbol attributes delivered through resource configuration, and Soft resources can depend on explicit parent availability or implicit DU/MT coexistence conditions. [5]

A more defensible formulation for your thesis is therefore to define a standardized availability indicator:

$$a_j[n] \in \{0, 1\}$$

derived from the IAB symbol state:

$$a_j[n] = \begin{cases} 1, & \text{Hard} \\ 1, & \text{Soft and available} \\ 0, & \text{Unavailable} \\ 0, & \text{guard symbol} \end{cases}$$

and then define the thesis-specific DU window as a maximal interval in which $a_j[n] = 1$.

That yields a direct distinction:

[A] Hard/Soft/Unavailable and guard-symbol restrictions come from IAB.

[C] The scalar interval $[t_j^{DU,start}, t_j^{DU,end}]$ is your derived simplification.

Your D2R constraint can then be written more rigorously as a research constraint over standardized availability:

$$\mathcal{R}_{j,k}^{D2R} \subseteq \{n : a_j[n] = 1\}.$$

The simpler form

$$t_{j,k}^{D2R,start} + T_{j,k}^{D2R} \leq t_j^{DU,end}$$

is valid if the available symbols are contiguous.

Energy harvesting and fairness

TS 38.300 only makes the RAN-level observation that an A-IoT device monitors R2D while it has sufficient energy; it does not standardize your Tag battery state equation or the exact harvested-energy conversion model. 3

Therefore the constraint on page 11 of your deck,

$$E_k^{bk}[\ell] + E_k^{EH}[\ell] \geq E_{\min},$$

is **[C] research modeling**, not a standards requirement. filecite turn0file0

Similarly, your objective

$$\min \left(\max_{k \in W[\ell]} B_k[\ell] - \min_{k \in W[\ell]} B_k[\ell] \right)$$

is **[C] your fairness policy**, not a required NR/IAB scheduler. The historical IAB study TR 38.874 considered scheduler/QoS and fairness questions, but a TR is a study report, not a normative protocol mandate. The final scheduler is free to use fairness, throughput, deadline, energy or another policy as long as normative resource and protocol constraints are respected. 43

That separation actually helps your thesis formulation:

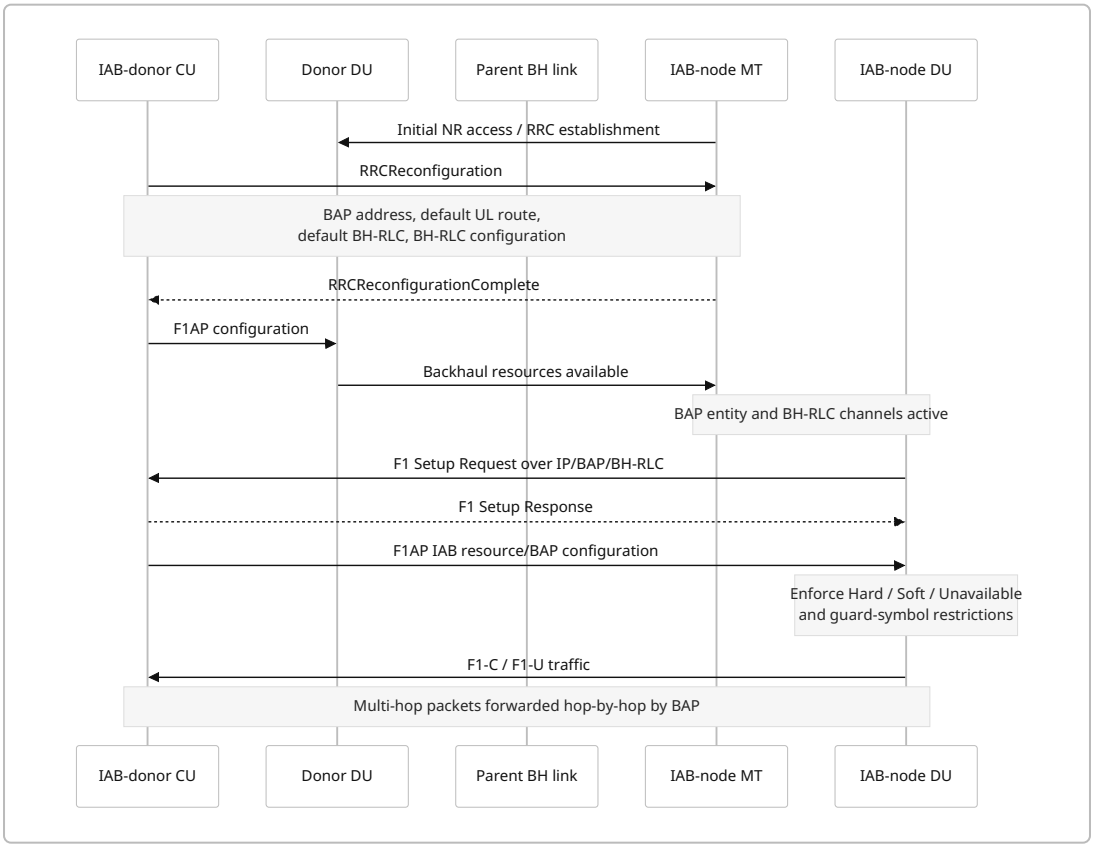
$$\boxed{\text{3GPP constraints}} + \boxed{\text{A-IoT protocol feasibility}} + \boxed{\text{energy model}} + \boxed{\text{fairness/deadline objective}}$$

rather than presenting the whole optimization problem as something prescribed by 3GPP.

Minimal protocol behaviors and message flows

For your MAC-level simulator, I would define a **Minimum Standards-Aligned Profile** rather than attempt every optional IAB feature.

The indispensable IAB behavior is: IAB-node integration, RRC/BAP bootstrap, BH-RLC establishment, BAP forwarding, F1 control/user traffic across one or more hops, and DU resource enforcement. 38.300 defines the architecture/protocol stack, 38.331 configures the IAB-MT/BAP state, 38.340 performs multi-hop BAP forwarding, and 38.401 gives the BH-RLC/F1 procedure relationships. 44

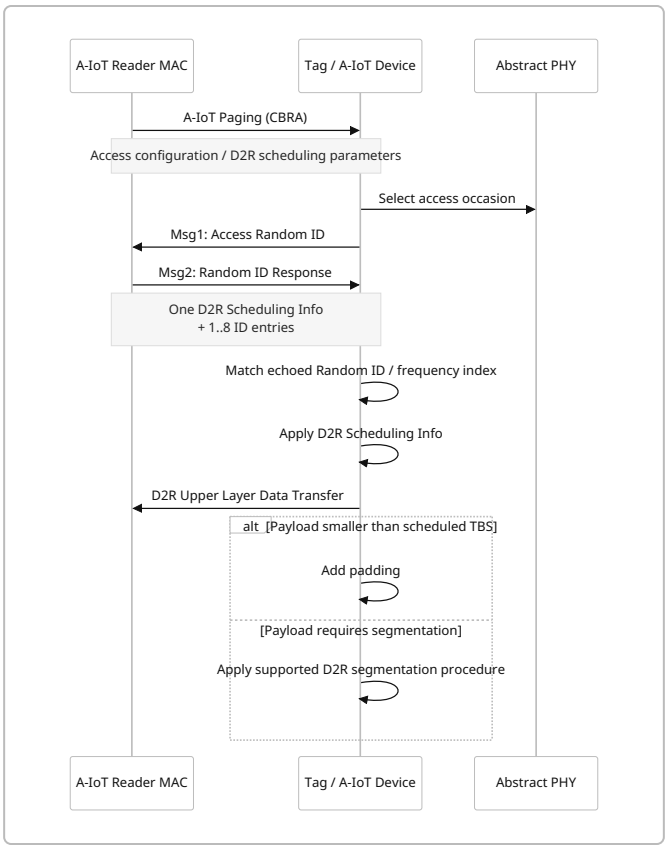


This sequence is a simulator-level condensation of the standardized architecture and procedures rather than a bit-exact trace. In particular, TS 38.401's BH-RLC establishment procedure contains explicit UE Context Modification and RRCReconfiguration exchanges. 16

A-IoT CBRA flow

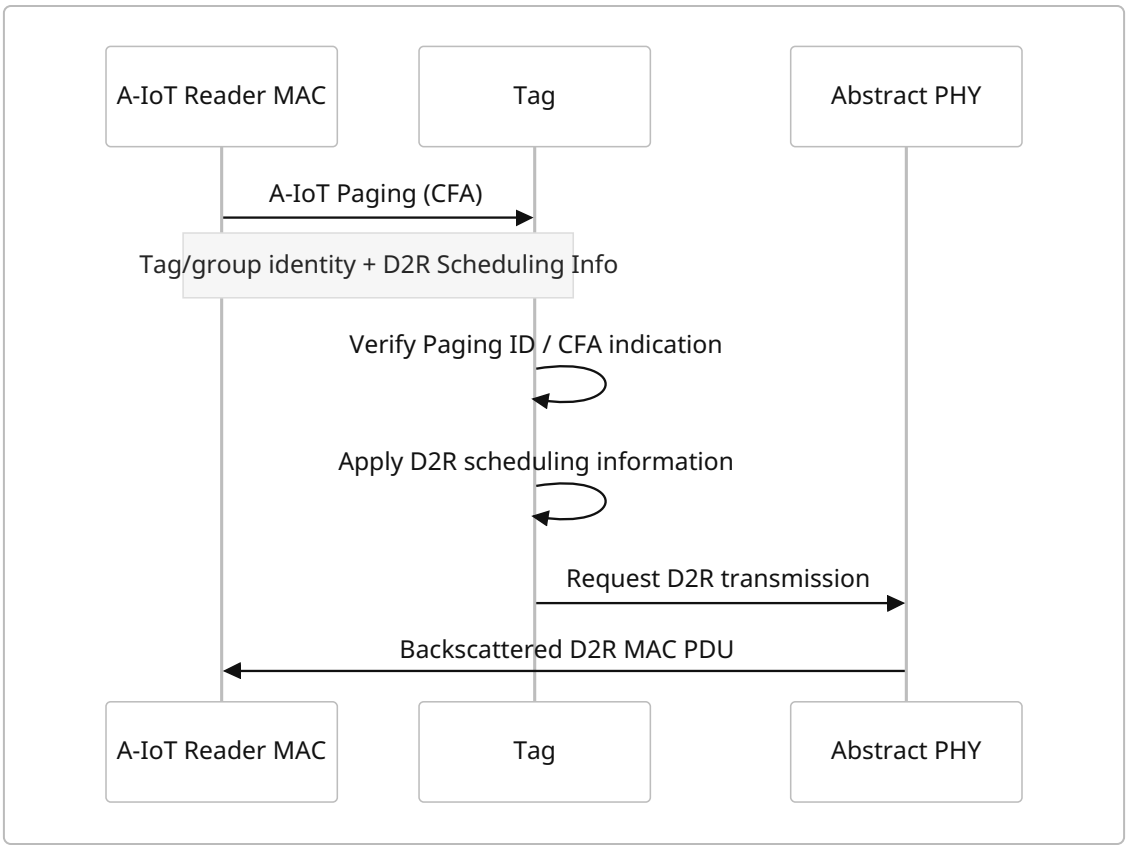
The CBRA path is important because it directly constrains how your Msg2 scheduler can assign TBS. After Paging indicates CBRA, the device selects an access occasion, transmits Msg1/Access Random ID, monitors

Random ID Response/Msg2, and after successful matching transmits D2R data using the resource supplied by Msg2. [45]



A-IoT CFA flow

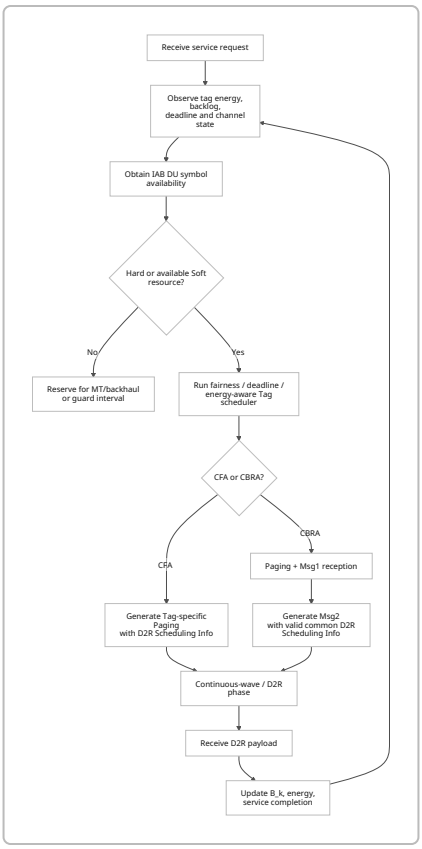
CFA is substantially simpler. 38.391 specifies that once CFA is initiated from Paging, the device proceeds to D2R transmission. [36]



This makes CFA particularly attractive for your thesis scheduler because you can associate a dedicated paging transmission with a selected Tag and therefore give that Tag a particular TBS without inventing a per-entry Msg2 extension.

Proposed IAB/A-IoT scheduling loop

The next flow is **[C] your research extension**, not a 3GPP architecture diagram:



It is a good thesis architecture provided you explicitly state that **placing the Reader function at the IAB-DU is the experimental extension**. Release-19 TS 38.300's A-IoT architecture does not define that split. 3

Minimum test suite

A useful protocol-level conformance/regression suite would test these behaviors:

Test	Expected behavior	Governing spec
IAB bootstrap	BAP entity/default route/default BH-RLC configured via RRC	38.331
BH-RLC establishment	CU/DU/RRC state transitions create matching BH channel	38.401/331/473
BAP one-hop forwarding	Routing ID selects expected next hop/BH-RLC	38.340
BAP multi-hop forwarding	Intermediate node forwards without terminating F1 payload	38.300/340
BAP congestion	Flow-control feedback influences route/link congestion state	38.340
Hard DU symbol	DU may schedule	38.300 §10.9

Test	Expected behavior	Governing spec
Unavailable symbol	DU scheduling rejected except specified exceptional cases	38.300 §10.9
Soft unavailable	DU scheduling rejected	38.300 §10.9
Soft available	DU scheduling permitted	38.300 §10.9
Guard interval	DU/MT transition consumes configured guard resource	38.300/321
CFA	Paging leads directly to D2R procedure	38.391
CBRA success	Msg1 ID matches Msg2 entry; D2R follows	38.391
CBRA mismatch	Msg2 is not accepted for the Tag	38.391
Msg2 1–8 IDs	Parser/state machine accepts legal list size	38.391
Multi-ID Msg2	All IDs share the Msg2 D2R Scheduling Info	38.391
Per-Tag TBS thesis mode	Separate grant/Paging/Msg2 is generated for each selected Tag	**C — thesis profile**
D2R short payload	MAC padding used	38.391
D2R long payload	Segmentation state exercised where specified	38.391
Energy insufficient	Suppress research-model transmission	**C — energy model**
Deadline/fairness selection	Objective affects selection but never violates 3GPP resource constraints	**C — scheduler**

Research evidence and implementation guidance

The standards deliberately leave the **scheduler objective**, detailed energy-harvesting model and network optimization algorithm open. That is where the academic literature is most useful.

IAB resource allocation and scheduling

"Freshness-Aware Energy Efficiency Optimization for Integrated Access and Backhaul Networks," IEEE Transactions on Wireless Communications, 2024. The work studies resource allocation in an IAB/IoT setting with a joint freshness/energy perspective and uses learning-based optimization, providing a useful precedent for treating IAB resource allocation as an optimization problem above the standardized protocol constraints rather than as a mandated scheduler. [46]

IEEE Xplore: document 10579786
Google Scholar: title search

Thesis use [B]: cite it when motivating the joint treatment of backhaul/access resource scarcity, energy cost and application-level freshness/deadline metrics.

“Toward Open Integrated Access and Backhaul with O-RAN,” IEEE MedComNet 2023, DOI 10.1109/MedComNet58619.2023.10168870. This work is useful on the implementation side because it examines how IAB control can be exposed in a programmable/open architecture rather than treating the network as an idealized relay chain. [47]

IEEE Xplore: document 10168870
Google Scholar: title search

Thesis use [B]: cite it to justify explicit controller/scheduler APIs and separation of topology/resource-management logic from packet forwarding.

A-IoT, backscatter and energy

“Challenges and Advances in Ambient IoT within 3GPP,” IEEE Communications Magazine, published online in 2025 and appearing in the 2026 volume, DOI **10.1109/MCOM.001.2500034**. It discusses the 3GPP A-IoT design problem, including extreme device-energy constraints, energy harvesting, random access and carrier-wave/backscatter issues. [48]

IEEE Xplore: document 11153767
Google Scholar: title search

Thesis use [B]: excellent motivation for your energy-aware Tag selection, but do not cite it as proof that 3GPP mandates your battery equation.

“State-of-the-Art Overview of 3GPP Ambient IoT Radio Access Aspects,” IEEE Communications Standards Magazine, 2025, DOI **10.1109/MCOMSTD.2025.3573815**. This is particularly useful for explaining the standardized A-IoT radio-access design around reader-to-device/device-to-reader operation and for placing 38.391 in the broader Release-19 context. [49]

IEEE Xplore: document 11016932
Google Scholar: title search

Thesis use [B]: good secondary academic explanation after you cite 38.300/38.391 themselves as the normative authority.

“Ambient IoT Devices in Future Cellular Networks: A System Level Study,” IEEE ICC Workshops 2024, DOI **10.1109/ICCWorkshops59551.2024.10615328**. The study examines system-level behavior of energy-constrained A-IoT devices and highlights the importance of device energy/sleep behavior and the timing relationship between harvesting and communication opportunities. [50]

IEEE Xplore: document 10615328
Google Scholar: title search

Thesis use [B]: directly relevant to your slot-level variable $E_k^{bk}[\ell]$ and energy-feasibility scheduling.

“Backscatter Communication Meets Practical Battery-Free Internet of Things: A Survey and Outlook,” IEEE Communications Surveys & Tutorials, 2023, DOI **10.1109/COMST.2023.3278239**, surveys practical battery-free/backscatter designs and the coupling between harvested energy, carrier-wave availability, link budget and achievable backscatter communication. [51]

IEEE Xplore: document 10130082
Google Scholar: title search

Thesis use [B]: use this to justify the **form** of an EH/backscatter energy model, while explicitly saying the numerical harvesting model is outside the 3GPP MAC requirements.

A useful citation pattern in the thesis would therefore be:

> “The protocol sequence and D2R grant structure follow 3GPP TS 38.391 [A], while the harvested-energy dynamics are modeled according to the backscatter/EH literature [B]. The proposed scheduler jointly selects Tags and scheduling resources subject to the 3GPP-derived availability constraints [C].”

That wording makes the standards/research boundary explicit.

Recommended simulator architecture and thesis claims

For an abstract-PHY implementation, the cleanest software architecture is approximately:

```
Simulator
│
├── Core / Service Generator
│   └── optional NGAP / AIOTF abstraction
│
├── IAB Donor
│   ├── CU
│   │   ├── RRC controller
│   │   ├── F1AP controller
│   │   └── optional XnAP / E1
│   └── Donor DU
│       ├── MAC scheduler
│       ├── BH RLC
│       └── BAP
│
├── IAB Node[j]
│   ├── IAB-MT
│   │   ├── RRC
│   │   ├── MAC
│   │   └── BH RLC / BAP
│   ├── IAB-DU
│   │   ├── F1 termination
```

```
│   │   ├── resource-availability scheduler
│   │   └── access MAC/RLC
│   │
│   └── Reader extension
│       └── A-IoT MAC 38.391
│           ├── Paging
│           ├── CFA
│           ├── CBRA Msg1/Msg2
│           └── D2R scheduler
│
├── Tag[k]
│   ├── A-IoT MAC state
│   ├── Energy state
│   └── Backscatter PHY abstraction
│
└── Research Scheduler
    ├── deadline metric
    ├── fairness metric
    ├── energy feasibility
    └── TBS / Tag-selection policy
```

The vertical division between `IAB-DU` and `Reader extension` is important. It lets you make a precise claim:

[A] the IAB-DU, IAB-MT, BAP, BH-RLC, F1 and IAB resource behavior follow 3GPP IAB specifications. 52

[A] Paging, CFA, CBRA, Msg2 and D2R MAC behavior follow the A-IoT specification TS 38.391. 53

[C] their colocation/interaction at a remote IAB node is the proposed research extension because Release-19 A-IoT explicitly excludes split-gNB support. 3

[C] energy harvesting, Tag-selection fairness, deadline-aware scheduling and the conversion of IAB symbol availability into a single DU scheduling window are optimization/modeling choices.

For the exact model in your uploaded slides, I would revise three pieces of terminology.

First, replace **“IAB-Node Reader” without qualification** by **“Proposed A-IoT Reader extension colocated with IAB-DU”** or equivalent. Your page-4 diagram already labels it as a research extension, which is the right direction. filecite turn0file0

Second, replace the statement that Msg2 provides an **individual Tag TBS field** with wording such as:

> “The scheduler determines a Tag-specific desired D2R TBS. The TBS is encoded through the D2R Scheduling Info of the Tag's applicable CFA Paging or Random ID Response. When independent TBS values are required for multiple Tags, the simulator schedules distinct

> grant-bearing R2D messages rather than assuming per-ID-entry TBS fields in a common Msg2."

That wording is consistent with the actual Msg2 structure in TS 38.391. [38]

Third, define your DU window as:

> "A simulator-level contiguous scheduling interval derived from the IAB-DU resources permitted by the Hard/Soft/Unavailable resource configuration, Soft-resource availability and required IAB guard symbols."

That formulation directly connects your deadline/D2R optimization to normative IAB scheduling constraints without claiming that `t_DU,start` and `t_DU,end` are standardized IEs. [5]

The resulting hierarchy for the thesis is rigorous:

$$\underbrace{\text{IAB topology} + \text{F1} + \text{BAP} + \text{BH-RLC} + \text{resource constraints}}_{\text{3GPP normative}}$$
$$+ \underbrace{\text{A-IoT Paging/CFA/CBRA/Msg2/D2R MAC}}_{\text{3GPP normative}}$$
$$+ \underbrace{\text{IAB-DU/Reader integration}}_{\text{research extension}}$$
$$+ \underbrace{\text{EH} + \text{fairness} + \text{deadline} + \text{per-Tag TBS optimization}.}_{\text{research model}}$$

That is the strongest standards-grounded framing for the simulator you are building: it preserves the protocol behavior that is actually standardized, exposes the places where 3GPP gives you scheduler freedom, and clearly identifies the novel IAB/A-IoT integration rather than inadvertently presenting it as an existing Release-19 requirement.

---

[1] [3] [5] [9] [10] [11] [12] [13] [44] [45] [52] TS 138 300 - V19.4.0 - 5G; NR; NR and NG-RAN Overall description; Stage-2 (3GPP TS 38.300 version 19.4.0 Release 19)
https://www.etsi.org/deliver/etsi_ts/138300_138399/138300/19.04.00_60/ts_138300v190400p.pdf

[2] [34] portal.3gpp.org
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[4] [35] [36] [37] [38] [39] [40] [53] TS 138 391 - V19.4.0 - 5G; NR; Ambient IoT Medium Access Control (MAC) protocol (3GPP TS 38.391 version 19.4.0 Release 19)
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ACTION_TYPE_NB=PU&FROM_DD=9&FROM_MM=Feb&FROM_YYYY=2026&REAL_FROM_DD=9&REAL_FROM_MM=Feb&REAL_FROM_YYYY=2026&REAL_TO_

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15 16 TS 138 401 - V19.3.0 - 5G; NG-RAN; Architecture description (3GPP TS 38.401 version 19.3.0 Release 19)
https://www.etsi.org/deliver/etsi_ts/138400_138499/138401/19.03.00_60/ts_138401v190300p.pdf

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https://portal.etsi.org/webapp/WorkProgram/Frame_WorkItemList.asp?SearchPage=TRUE+++&butExpertSearch=++Search++&includeNonActiveTB=FALSE+++&optDisplay=ALL+++&qCURRENT_STATE_CODE=finalized+++&qINCLUDE_SUB_TB=True+++&qPROJECT_BOOLEAN=OR+++&qREPORT_TYPE=SUMMARY+++&qSORT=HIGHVERSION+++&qTITLE=TS+38.470+++&titleType=all+++&utm_source=chatgpt.com

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https://portal.etsi.org/webapp/WorkProgram/Frame_WorkItemList.asp?SearchPage=TRUE+++&butExpertSearch=++Search++&includeNonActiveTB=FALSE+++&optDisplay=ALL+++&qCURRENT_STATE_CODE=finalized+++&qINCLUDE_SUB_TB=True+++&qPROJECT_BOOLEAN=OR+++&qREPORT_TYPE=SUMMARY+++&qSORT=HIGHVERSION+++&qTITLE=TS+38.423+++&titleType=all+++&utm_source=chatgpt.com

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